

TYLER P. QUIGLEY, PH.D.

Life Sciences | AI-Accelerated Research | Scientific Strategy

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Research scientist with deep expertise in experimental biology, quantitative analysis, and AI-augmented scientific workflows. Proven ability to design and execute rigorous research programs, translate complex biological insights into actionable strategies, and bridge computational approaches with wet-lab realities. Experienced in evaluating research programs, assessing scientific claims, and building systems that support interpretable, high-quality science at scale.

EDUCATION

Ph.D., Animal Behavior

Arizona State University | Graduated April 2024

Dissertation: *"The Structure and Function of the Honeybee Blood-Brain Barrier"*

B.S., Biology

Roanoke College | Graduated 2015

PROFESSIONAL EXPERIENCE

Science Lead

PsyDAO | FT, Remote | Jan 2025 - Present

- Lead scientific strategy and research evaluation for decentralized biotech initiatives in neuroscience, psychedelics, and longevity biology
- Design AI-augmented workflows for literature synthesis, experimental planning, and hypothesis generation while maintaining epistemic rigor and identifying model limitations
- Evaluate research proposals for experimental design quality, statistical validity, mechanistic plausibility, and safety/misuse risks
- Translate complex biological mechanisms into strategic frameworks for non-expert stakeholders

IP Technical Specialist, Life Sciences

Calyx Law | FT, Remote | Apr 2024 - Dec 2024

- Evaluated biotech inventions for commercial viability, technical feasibility, and competitive positioning
- Conducted prior art searches and analyses for patent applications in psychedelics, AI therapeutics, and emerging biotechnologies
- Translated complex scientific mechanisms into technical documentation for patent drafting and legal audiences
- Collaborated with inventors and legal teams on patentability assessment and strategic IP positioning

Technology Transfer Fellow

Skysong Innovations, Arizona State University | PT, Scottsdale, AZ | Apr 2023 - Aug 2023

- Evaluated life sciences inventions for technical merit, market potential, and commercial pathways
- Organized and analyzed IP data using management software to streamline technology transfer decisions
- Facilitated communication between inventors, researchers, and commercialization teams

IP Research Assistant

Calyx Law | PT, Remote | Mar 2021 - Apr 2023, Sept 2023 - Apr 2024

- Supported patent applications in psychedelics, AI, and web3 technologies through targeted research and prior art analysis
- Synthesized complex scientific literature into research reports informing patent strategy and prosecution
- Identified novel uses and compositions across emerging therapeutic modalities to support differentiation arguments

Doctoral Researcher, Animal Behavior & Neuroscience

Arizona State University, School of Life Sciences | Tempe, AZ | Aug 2015 - Apr 2024

- Pioneered quantitative methods to characterize blood-brain barrier ultrastructure in an invertebrate model, utilizing novel stereological imaging workflows (TEM, SEM, array tomography, NanoSIMS) where established protocols did not exist
- Development novel assays, sample preparation and imaging protocols to assess function of nanoscale structures, systematically troubleshooting and iterating to achieve reproducible results
- Secured over \$20,000 in competitive grants by articulating novel questions and justifying unconventional methodological approaches
- Mentored six undergraduate researchers on experimental design, troubleshooting, and developing solutions when standard methods fail

SELECTED PUBLICATIONS

Quigley, T.P., & Amdam, G.V. (2024). Ultrastructural Organization of the Honeybee Blood-Brain Barrier and Comparison with Age. *BioRxiv*. In review. DOI: 10.1101/2024.09.27.615080

Quigley, T.P., & Amdam, G.V. (2024). Effect of aging and Varroa parasitism on paracellular and transcellular permeability of the honeybee blood-brain barrier. *BioRxiv*. In review. DOI: 10.1101/2024.09.27.615259

Quigley, T.P., & Amdam, G.V. (2021). Social modulation of ageing: mechanisms, ecology, evolution. *Philosophical Transactions of the Royal Society B*, 376(1823), 20190738.

Quigley, T.P., Amdam, G.V., & Harwood, G. (2019). Honey bees as bioindicators of changing global agricultural landscapes. *Current Opinion in Insect Science*, 35, 132-137.

TECHNICAL SKILLS

Experimental Biology: Advanced electron microscopy (TEM, SEM, serial/array tomography), confocal microscopy, immunolabeling, NanoSIMS, sample preparation (chemical/cryo fixation, ultramicrotomy), stereological analysis, microdissection

Computational & Analysis: R programming (tidyverse, ggplot2), quantitative image analysis (ImageJ, NanoSIMS workflows), statistical modeling, experimental design

AI-Augmented Research: LLM-assisted literature synthesis, hypothesis generation, structured scientific reasoning, evaluation methodology design, identifying model limitations and failure modes

Scientific Communication: Technical writing, grant proposal development, IP assessment and patent analysis, cross-disciplinary translation, research evaluation and peer review

Domain Expertise: Neuroscience, physiology, aging biology, blood-brain barrier function, psychedelic science, biotechnology evaluation

SCIENCE COMMUNICATION & OUTREACH

The Tab Psychedelic Science Newsletter | Self-published | Mar 2021 - Present | 800+ subscribers

Weekly newsletter translating psychedelic research into accessible analysis for diverse audiences, covering molecular mechanisms, clinical trials, and ethical considerations

Freelance Science Writer | 2021 - Present

Articles and reviews on bioactivity of pharmaceutical compounds, research methodologies, and emerging therapeutic applications

Ask-A-Biologist Contributor | Arizona State University | 2015 - 2018

Created educational content translating research for K-12 and general audiences